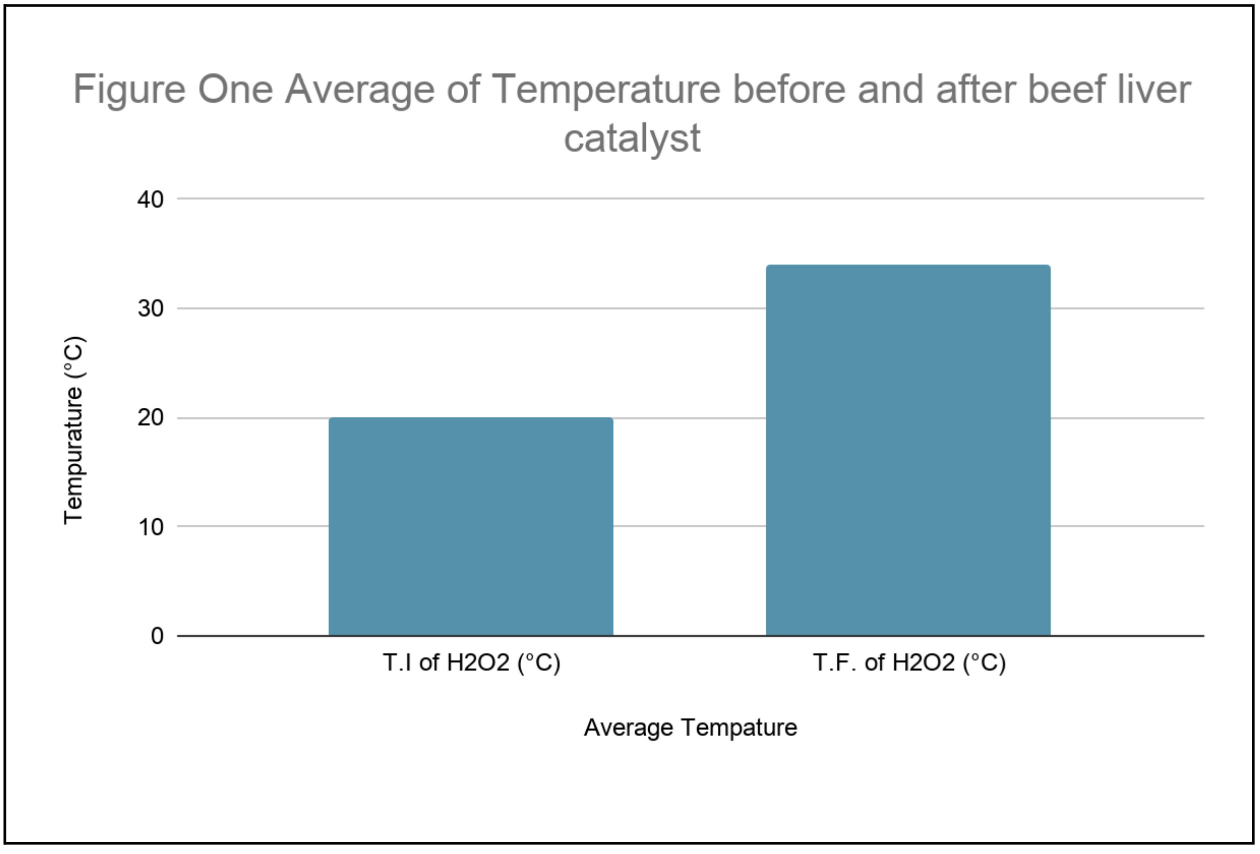
Results and Analysis:

- ☑ ~~Must contain a **complete set of quantitative data (raw data from three trials of the experiment and average data)** in a data table with table number, descriptive title, and headings, units and quantitative data in the correct places~~
- ☑ ~~Must contain a **graph of the average data that compares the change in temperature before and after the reaction** with the x and y-axis labelled with units, a figure number and descriptive title (you must decide the appropriate type of graph to use for your data)~~
- ☑ ~~Must contain a summary of the trends shown in the data collected~~

Insert data table(s) in the box below:

Trial	Trial 1	Trial 2	Trial 3	Average
T.I of H_2O_2 (°C)	20°C	20°C	20°C	20°C
T.F of H_2O_2 (°C)	32°C	34°C	36°C	34°C

Insert graph(s) in the box below:



Summary of trends in the data:

- ☑ Accurately interpret data and **describe** results using correct scientific reasoning

The temperature of the hydrogen peroxide consistently increased by an average of 14°C. The starting temperature of the hydrogen peroxide was consistently 20°C, with an increase of at least 12°C and at most 16°C.

Comparison, Analysis, and Conclusion:

- ☑ ~~Accepts or rejects the hypothesis while re-stating the hypothesis~~
- ☑ ~~**Discuss**, by comparing the data collected on all variables, why the hypothesis was proven correct or incorrect~~
- ☑ ~~**Draw a conclusion** based on the data - did the temperature change or not and did this indicate a chemical change or not~~
- ☑ ~~Must use valid sources and must have valid MLA **in-text citations** for any sources used - **your document must contain at least two different valid sources**~~

I reject the hypothesis stating that "if a catalyst of beef liver is added to hydrogen peroxide, then the temperature of the hydrogen peroxide will drop, indicating that a chemical change has happened, because the beef liver will absorb heat from the hydrogen peroxide, causing it to become colder."

The corrected hypothesis is:
"If a catalyst of beef liver is added to hydrogen peroxide, then the temperature of the hydrogen peroxide will RISE, indicating that a chemical change has happened, because the beef liver will RELEASE HEAT, causing the hydrogen peroxide to become WARMER."

Table 1 and Figure 1 support this conclusion. The temperature of the hydrogen peroxide rose in all the experiments, and its average increase was about 14°C. As the only factor added was the beef liver, it shows that the temperature rise was due to the reaction of catalase from the liver breaking down hydrogen peroxide into water and oxygen (NCBI).

This temperature rise proves that there was a chemical reaction. The decomposition of hydrogen peroxide by catalase results in the formation of water and oxygen, accompanied by the release of heat. Therefore, the process releases heat to its surroundings (Science Learning Hub).

In conclusion, the temperature increased, showing that a chemical change had happened. Therefore, the experimental data support the corrected hypothesis.

Evaluation of Method:

- ☑ ~~**State** two sources of experimental error - these are errors that are embedded into the method of your experiment~~
- ☑ ~~**Discuss** the problem caused by this error with respect to the data collection~~
- ☑ ~~**Discuss** ways to fix each error for future experiments~~

Error 1:

Source of Error:	**The Problem(s) caused by this error:**	**Way to fix the error**
The thermometer	**It could still be warm in trials 2 & 3 from the previous trial**	**A way of fixing this error would be using separate thermometers for each trial**

Error 2:

Source of Error:	**The Problem(s) caused by this error:**	**Way to fix the error**
The size and weight of the beef liver pieces	**It could lead to a different result if the beef liver wasn't equal in all three trials**	**Measuring each individual piece of beef liver making sure they are all the same weight and size.**

MLA Works Cited:

- ☑ ~~Must contain **at least two** relevant sources~~
- ☑ ~~Must be alphabetized by last name of the author of first letter of article title; whichever you have~~
- ☑ ~~Anything cited in the Works Cited must also be cited in-text~~
- ☑ ~~Must follow MLA format~~

"Catalase." *PubChem*, National Center for Biotechnology Information (NCBI), U.S. National Library of Medicine, https://pubchem.ncbi.nlm.nih.gov/compound/Catalase. Accessed 10 May 2026.

“Enzymes and Catalysts.” *Science Learning Hub*. University of Waikato, https://www.sciencelearn.org.nz/resources/1549-enzymes-and-catalysts. Accessed 10 May 2026.